

COURSE INFORMATION SHEET
Programming for Problem Solving Lab

Course Code: CS24102

Course Title: Programming for Problem Solving Lab

Pre-requisite(s): School level Mathematics and Science

Co- requisite(s): Programming for Problem Solving

Credits: L: 0 T: 0 P:2

Class schedule per week: 2

Class: UG

Semester / Level: II

Branch: ALL

COURSE OBJECTIVES

This course enables the students to:

1.	Compile, write and debug simple programs with sequential logic.
2	Apply selection and iteration constructs to solve real-world problems.
3	Explore the use arrays and string handing in implementing solutions to problems.
4	Design and develop solutions to problems using modular programming constructs such as functions.
5	Explore the use of structures and pointers to solve real-world problems.

COURSE OUTCOMES (COs)

After the completion of this course, students will be able to:

CO1	Formulate Algorithms and draw flowchart for arithmetic and logical problems.
CO2	Translate the algorithms/flowcharts to programs.
CO3	Test and execute the programs and correct syntax and logical errors.
CO4	Decompose a problem into functions and develop modular reusable code.
CO5	Use arrays, pointers, strings and structures to solve real-world problems.

SYLLABUS

MODULE	(NO. OF LECTURE HOURS)
Module – I 1. Learn how to compile and debug C programs. (preference should be given to UNIX environments.) 2. Write simple programs using sequential logic requiring the declaration of variables for data types. 3. Write programs that produce formatted output. 4. Write programs to acquire data from users and use them in their programs.	4
Module – II 1. Write simple programs to understand the workings of selection structures. 2. Write programs using concepts of nested selection structures. 3. Learn how to program the switch case structure.	4
Module – III 1. Write programs using different iterative structures. 2. Write programs using nested iterative structures. 3. Write programs that embed selection structures in loops and vice versa. 4. Writing programs to create and use arrays. 5. Write programs to manipulate strings	4
Module – IV 1. Write simple functions demonstrating the concepts of parameter passing and return values. 2. Write programs to access global/extern variables from functions. 3. Write programs to demonstrate calling functions from functions.	6
Module – V 1. Write programs to create and use structures. 2. Write programs to demonstrate the concept of array of structures and passing structure variables to functions. 3. Writing programs to create pointers and understanding their basic properties. 4. Performing call by reference function calls.	8

TEXTBOOKS:

- Let us C, Yashwant Kanetkar, 18th Edition, BPB Publications

- Byron Gottfried, Schaum's Outline of Programming with C, McGraw-Hill
- E. Balaguruswamy, Programming in ANSI C, Tata McGraw-Hill
- R.G.Dromey, How to Solve it by Computer, Pearson Education

REFERENCE BOOKS:

- Brian W. Kernighan and Dennis M. Ritchie, The C Programming Language, Prentice.

GAPS IN THE SYLLABUS (TO MEET INDUSTRY/PROFESSION REQUIREMENTS)

- The syllabus focused on the concepts and basics of Program writing skills.
- Industry often requires debugging of their existing programs/software compared to the new program, which is a knowledge beyond the basics, including real-world software (collection of programs) experience.
- More memory management practices, file handling and library functions

POS MET THROUGH GAPS IN THE SYLLABUS: YES [PO1-PO5 & PO10-PO12]

TOPICS BEYOND SYLLABUS/ADVANCED TOPICS/DESIGN:

File Handling with memory management, preprocessor directives, Graphics, Data Arrangement, Task scheduling and assembly level programs.

POS MET THROUGH TOPICS BEYOND SYLLABUS/ADVANCED TOPICS/DESIGN: YES [PO1-PO5]

COURSE OUTCOME (CO) ATTAINMENT ASSESSMENT TOOLS & EVALUATION PROCEDURE

DIRECT ASSESSMENT

Assessment Tool	% Contribution during CO Assessment
Continuous Internal Assessment	60
End Semester Examination	40

Continuous Internal Assessment	% Distribution
Day-to-day performance & Lab files	30
Quiz	10
Viva	20

End Semester Examination	% Distribution
Examination Experiment Performance	30
Quiz	10

Assessment Components	CO1	CO2	CO3	CO4	CO5
Continuous Internal Assessment	20%[12]	20%[12]	20%[12]	20%[12]	20%[12]
Semester End Examination	20%[8]	20%[8]	20%[8]	20%[8]	20%[8]

INDIRECT ASSESSMENT

4. Student Feedback on Course Outcome
5. Student Feedback on Faculty/Content Delivery
6. Student Feedback on Evaluation Procedures

COURSE DELIVERY METHODS

CD1	Demonstration by use of smart boards/LCD projectors
CD2	Assignments
CD3	Viva-Voce/Quiz (s)
CD4	Software and Hardware
CD5	Laboratory experiments/Coding

MAPPING BETWEEN COURSE OUTCOMES AND POs and PSOs

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO 2	PSO 3
CO1	3	3	3	3	2	1	0	0	1	2	1	1	2	3	2
CO2	3	3	3	3	2	1	0	0	2	1	1	2	2	3	2
CO3	3	3	3	3	2	1	0	0	1	2	1	2	3	3	2
CO4	3	3	3	3	2	2	0	0	2	2	2	2	3	2	2
CO5	3	2	2	1	2	2	0	0	1	2	2	2	3	3	3

Grading: No correlation – 0, Low correlation - 1, Moderate correlation – 2, High Correlation - 3

MAPPING BETWEEN COURSE OUTCOMES AND COURSE DELIVERY METHOD

Course Outcomes	Course Delivery Method
CO1	CD1, CD2, CD3
CO2	CD3, CD5
CO3	CD3, CD5
CO4	CD2, CD3, CD4
CO5	CD1, CD3, CD5